

① Drive (laser  $a_0/\tau$  or RF cavity)ORANGE – wet-lab driver



② Plasma / cavity ( $n_p$  response)GREEN/BLUE – cold-linear



③ Wakefield  $E$ -field (Dawson  $E_0$ )GREEN/BLUE – gate #1007



④  $e^-$  acceleration (1-D linear PIC)GREEN – #1088  $\Delta$  3.56%



⑤ Focusing (FODO twiss, Xsuite)GREEN – rel  $10^{-6}$



⑥ Energy loss (Bethe-Bloch  $dE/dx$ )GREEN/BLUE – relerr  $4e-10$



⑦ Event analysis (pylhe LHE stats)ORANGE – absorbed=false